

Problem Statement

Evaluate the limit presented in the image “dailyintegral-limit-limit-1x1-2026-07-06.jpg”:

$$\lim_{x \rightarrow \infty} (1 + x^{-3/2} + x^{-2})^x$$

Detailed Solution

This limit initially presents an indeterminate form of type 1^∞ . A standard and elegant way to evaluate limits of this form is to use the natural logarithm to bring the variable exponent down, evaluate the resulting limit, and then exponentiate the final answer.

Step 1: Apply the natural logarithm

Let L represent the value of the limit:

$$L = \lim_{x \rightarrow \infty} (1 + x^{-3/2} + x^{-2})^x$$

By taking the natural logarithm of both sides, we can use the power rule for logarithms ($\ln(a^b) = b\ln(a)$) and the continuity of the logarithm function to move the limit outside:

$$\begin{aligned}\ln(L) &= \ln \left(\lim_{x \rightarrow \infty} (1 + x^{-3/2} + x^{-2})^x \right) \\ &= \lim_{x \rightarrow \infty} \ln \left((1 + x^{-3/2} + x^{-2})^x \right) \\ &= \lim_{x \rightarrow \infty} x \ln (1 + x^{-3/2} + x^{-2})\end{aligned}$$

Step 2: Evaluate using series expansion

Notice that as $x \rightarrow \infty$, the expression $u = x^{-3/2} + x^{-2}$ approaches 0. Because $u \rightarrow 0$, we can utilize the Maclaurin series expansion for $\ln(1 + u)$:

$$\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \dots$$

Substituting $u = x^{-3/2} + x^{-2}$ into this expansion yields:

$$\begin{aligned}\ln(1 + x^{-3/2} + x^{-2}) &= (x^{-3/2} + x^{-2}) - \frac{1}{2} (x^{-3/2} + x^{-2})^2 + \mathcal{O}(x^{-4.5}) \\ &= x^{-3/2} + x^{-2} - \frac{1}{2} (x^{-3} + 2x^{-7/2} + x^{-4}) + \dots\end{aligned}$$

Now, we substitute this expanded form back into our limit for $\ln(L)$ and distribute the x multiplier:

$$\begin{aligned}
\ln(L) &= \lim_{x \rightarrow \infty} x \left[x^{-3/2} + x^{-2} - \frac{1}{2}x^{-3} - \dots \right] \\
&= \lim_{x \rightarrow \infty} \left[x(x^{-3/2}) + x(x^{-2}) - x \left(\frac{1}{2}x^{-3} \right) - \dots \right] \\
&= \lim_{x \rightarrow \infty} \left[x^{-1/2} + x^{-1} - \frac{1}{2}x^{-2} - \dots \right]
\end{aligned}$$

Step 3: Compute the final limit

As x grows infinitely large ($x \rightarrow \infty$), any term of the form x^{-p} where $p > 0$ will converge to 0. Therefore:

$$\lim_{x \rightarrow \infty} \frac{1}{x^{1/2}} = 0, \quad \lim_{x \rightarrow \infty} \frac{1}{x} = 0, \quad \text{etc.}$$

Evaluating the limit term by term, all terms vanish:

$$\ln(L) = 0 + 0 - 0 - \dots = 0$$

We have found that $\ln(L) = 0$. To find our original limit L , we simply exponentiate both sides with base e :

$$L = e^0 = 1$$

Final Answer:

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